Chaohao Yang

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Personal Website

Education

- **Johns Hopkins University** *Aug. 2025 - Expected May 2027*
  Master of Science in Engineering in Computer Science GPA: 3.79/4.00
  – Core Coursework: Natural Language Processing, Deep Learning, Learning-based Vision
- **The Chinese University of Hong Kong, Shenzhen (CUHKSZ)** *Sept. 2021 - Jun. 2025*
  Bachelor of Engineering in Computer Science and Engineering GPA: 3.74/4.00
  – Core Coursework: Python, C++, Calculus, Linear Algebra, Data Structures, Computer Architecture, Operating System, Database, Parallel Programming, Software Engineering
- **University of California, Berkeley** *Jan. 2024 - May 2024*
  GLOBE Visiting Student Program GPA: 4.00/4.00
  – Core Coursework: Algorithm, Machine Learning, Artificial Intelligence

Publications

- Han, Y.*, **Yang, C.***, Chen, C., Wang, X., & Sun, R. (2025). Q-Adam-mini: Memory-efficient 8-bit quantized optimizer for large language model training. In *ICML 2025 Workshop ES-FoMo-III.*
  [co-first author]
- **Yang, C.**, & Ding, C. (2024). Learning word embedding with better distance weighting and window size scheduling. https://doi.org/10.48550/arXiv.2404.14631
- Han, R.*, **Yang, C.***, Peng, T., Tiwari, P., Wan, X., Liu, L., & Wang, B. (2024). An empirical study on information extraction using large language models. https://doi.org/10.48550/arXiv.2305.14450
  [co-first author]

Research Experience

- **Johns Hopkins University (Bloomberg Distinguished Professor Alan Yuille)** *Jan. 2026 - Jul. 2026*
  *A unified framework for controllable video generation with **world-centric 4D guidance***
  – Built a unified evaluation interface for controllable video generation, integrating **PSNR**, **LPIPS**, and camera trajectory metrics to compare **Wan**, **WanMove**, **ATI**, and our model
  – Standardized evaluation of foreground motion and camera motion under a common geometry-aware protocol, supporting fair analysis of sparse and dense guidance settings
  – Planning to submit to *ICLR 2027*
- **CUHKSZ (Associate Professor Ruoyu Sun)** *Sept. 2024 - May 2025*
  *Developing a quantized version for **Adam-mini** optimizer*
  – Pre-trained and fine-tuned Llama models (**60M-8B** parameters) on the **C4**, **MMLU**, and **GSM-8K** datasets using AdamW, Adam-mini, or **Low-Rank Adaptation (LoRA)**, as experimental baselines
  – Developed the quantized Adam-mini, performing specialized quantization schemes on different parameters, and adopting **stochastic rounding** to ensure effective optimizer updates
  – Compared the performance of the quantized Adam-mini with baselines under the same settings, showed that the quantized Adam-mini achieved comparable performance with only **15%** of the optimizer state memory of AdamW
- **CUHKSZ (Presidential Chair Professor Chris Ding)** *Jun. 2023 - Jan. 2024*
  *Introducing distance information into the **Word2Vec** model*
  – Put forward the epoch-based dynamic window size strategy for **Skip-gram** to sample more from closer context words in a more stable manner.
  – Conducted the formulated learnable distance-related weights for the average pooling of **CBOW**, taking both the modeling effect and adaptability into account
  – Achieved **15.3%** accuracy improvement for CBOW and **2.5%** for Skip-gram on the Google analogy test set, methods proven effective
- **CUHKSZ (Assistant Professor Benyou Wang)** *Dec. 2022 - Jun. 2023*
  *Evaluating the information extraction ability of **large language models (LLMs)***
  – Collected and cleaned **over 20** high-quality information extraction task datasets to evaluate the information extraction ability of LLMs including GPT-4 and GPT-3.5, of which **16** datasets were selected for experiments
  – Designed multi-threaded Python programs to efficiently send information extraction prompts to LLMs and collect responses, then monitored the program's execution
  – Calculated **precision**, **recall**, and **F1** for the information extraction ability of LLMs as evaluation metrics based on the collected responses

Internship Experience

- **Tencent TiMi Studio Group** *May 2024 - Aug. 2024*

  Performance R&D Group Intern

  – Refined the term tables for Chinese-to-English, English-to-French, and English-to-German translations to support the overseas release of Tencent's **Honor of Kings (HOK)** and **Delta Force (DF)**
  – Successfully trained translation models for the two games using Llama-3-70B, enhancing automation and efficiency in language localization for their international launches
  – Introduced the **retrieval-augmented generation (RAG)** technique to the models to address term translation challenges and improve model performance, achieving **translation edit rates (TER) below 15** for HOK and **below 10** for DF, in line with company targets

Campus Activities

- **CUHKSZ Mathematical Modeling Club** *Nov. 2022 - Jun. 2025*

  Committee Member

  – Advised for the detailed process of club activities, such as site selection and participant guidance
  – Offered information technology support for activity logistics, including automatic arrangements of personnel and activity-related software maintenance

- **CUHKSZ Graduate course: Artificial Intelligence** *Sept. 2023 - Dec. 2023*

  Undergraduate Student Teaching Fellow

  – Designed **5 homework projects** for the course, covering multiple aspects including word embedding and pre-trained language models, and scored the project completion of all **92 students** in the course
  – Provided students with **10 tutorials** about the implementation details of AI models covered in lectures

Awards

- **Dean's List Certificates** for excellent academic performance *academic years from 2021 to 2025*
- **Undergraduate Research Award** for excellent research achievements *2023*
- **Student Union Certificate of Honor** for contributions to the student club *academic year 2022-2023*

Technical Skills

Programming Languages: Python, C/C++, SQL, Bash
AI/ML: PyTorch, CUDA, DeepSpeed, Hugging Face Libraries, OpenAI API
Tools & Libraries: Git, Linux, LaTeX, Weights & Biases, NumPy, Pandas, NLTK